**Q1 - 25 July - Shift 1**

Three identical particle A, B and C of mass 100 kg each are placed in a straight line with AB = BC = 13 m. The gravitational force on a fourth particle P of the same mass is F, when placed at a distance 13 m from the particle B on the perpendicular bisector of the line AC. The value of F will be approximately :

(A) 21 G (B) 100 G

(C) 59 G (D) 42 G

*Space for your notes:*

**Q2 - 25 July - Shift 2**

An object is taken to a height above the surface of earth at a distance $\frac{5}{4}R$ from the centre of the earth. Where radius of earth, $R = 6400$ km. The percentage decrease in the weight of the object will be

(A) 36% (B) 50%

(C) 64% (D) 25%

*Space for your notes:*

**Q3 - 26 July - Shift 1**

The percentage decrease in the weight of a rocket, when taken to a height of 32 km above the surface of earth will, be :

(Radius of earth = 6400km)

(A) 1 % (B) 3%

(C) 4% (D) 0.5%

*Space for your notes:*

**Q4 - 26 July - Shift 2**

A body is projected vertically upwards from the surface of earth with a velocity equal to one third of escape velocity. The maximum height attained by the body will be:
(Take radius of earth = 6400 km and g=10 $ms^{-2}$ )

(A) 800 km (B) 1600 km

(C) 2133 km (D) 4800 km

*Space for your notes:*

**Q5 - 27 July - Shift 1**

Two satellites A and B having masses in the ratio 4:3 are revolving in circular orbits of radii 3r and 4 r respectively around the earth. The ratio of total mechanical energy of A to B is :

(A) 9 : 16 (B) 16 : 9

(C) 1 : 1 (D) 4 : 3

*Space for your notes:*

**Q6 - 27 July - Shift 2**

A body of mass m is projected with velocity $\lambda v_e$ in vertically upward direction from the surface of the earth into space. It is given that $v_e$ is escape velocity and $\lambda < 1$. If air resistance is considered to the negligible, then the maximum height from the centre of earth, to which the body can go, will be (R : radius of earth)

(A) $\frac{R}{1+\lambda^2}$ (B) $\frac{R}{1-\lambda^2}$

(C) $\frac{R}{1-\lambda}$ (D) $\frac{\lambda^2 R}{1-\lambda^2}$

*Space for your notes:*

**Q7 - 28 July - Shift 1**

If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately :

(A) decrease by 2% (B) decrease by 4%

(C) increase by 2% (D) increase by 4%

*Space for your notes:*

**Q8 - 28 July - Shift 2**

Assume there are two identical simple pendulum Clocks-1 is placed on the earth and Clock-2 is placed on a space station located at a height h above the earth surface. Clock-1 and Clock-2 operate at time periods 4s and 6s respectively. Then the value of h is –

(consider radius of earth $R_E$ = 6400 km and g on earth 10 $m/s^2$)

(A) 1200 km

(B) 1600 km

(C) 3200 km

(D) 4800 km

*Space for your notes:*

**Q9 - 29 July - Shift 1**

If the acceleration due to gravity experienced by a point mass at a height h above the surface of earth is same as that of the acceleration due to gravity at a depth $\alpha h$ ($h \ll R_e$) from the earth surface. The value of $\alpha$ will be __________.

(use $R_e$ = 6400 km)

*Space for your notes:*

**Q10 - 29 July - Shift 2**

An object of mass 1 kg is taken to a height from the surface of earth which is equal to three times the radius of earth. The gain in potential energy of the object will be

[If, g=10$ms^{-2}$ and radius of earth = 6400 km]

(A) 48 MJ (B) 24 MJ

(C) 36 MJ (D) 12 MJ

*Space for your notes:*



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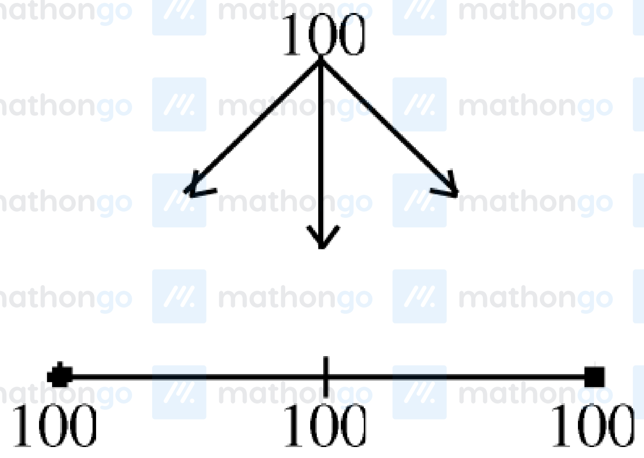
# Answer Key

| | | | |
|---|---|---|---|
| **Q1** (B) | **Q2** (A) | **Q3** (A) | **Q4** (A) |
| **Q5** (B) | **Q6** (B) | **Q7** (D) | **Q8** (C) |
| **Q9** (2) | **Q10** (A) | | |

**Q1 (B)**



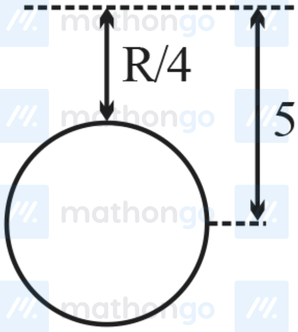

$$F=\frac{GMM}{r^2}+\sqrt{2}\frac{GMM}{(\sqrt{2}r)^2}$$

$$=\frac{GMM}{r^2}\left(1+\frac{1}{\sqrt{2}}\right)$$

$$=\frac{G\times10^4}{13^2}\left(1+\frac{1}{\sqrt{2}}\right)$$

$$F\simeq100G$$

**Q2 (A)**




$$g_{eff}=\frac{g}{\left(1+\frac{h}{R}\right)^2};\ g_{eff}=\frac{g}{\left(1+\frac{1}{4}\right)^2}=\frac{16g}{25}$$

$$\text{change}=\frac{g_{eff}-g}{g}\times100=\frac{\frac{16}{25}-1}{1}\times100$$

$$=\frac{-9}{25}\times100=-36\%$$

Hence % decrease in the weight = 36%

**Q3 (A)**

Acceleration due to gravity at a height h<< R is

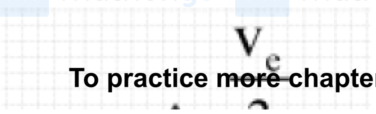
$$g'=g\left(1-\frac{2h}{R}\right)$$

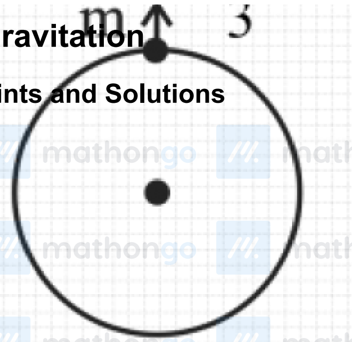
$$\therefore\ \frac{\Delta g}{g}=\frac{2h}{R}$$

$$\Rightarrow\ \frac{\Delta g}{g}\times100=\frac{2h}{R}\times100$$

$$=2\times\frac{32}{6400}\times100=1\%$$

**Q4 (A)**






$$V_e = \sqrt{\frac{2Gm}{R}}$$

$$\frac{-GMm}{R} + \frac{1}{2}m\frac{V_e^{\,2}}{9} = -\frac{GMm}{R+h}$$

$$\frac{GM}{R+h} = \frac{GM}{R} - \frac{V_e^{\,2}}{18}$$

$$\frac{GM}{R+h} = \frac{GM}{R} - \frac{GM}{9R}$$

$$\frac{GM}{R+h} = \frac{8GM}{9R}$$

$$\frac{1}{R+h} = \frac{8}{9R}$$

$$9R = 8R + 8h$$

$$h = \frac{R}{8} \Rightarrow \frac{6400}{8} \Rightarrow 800\text{km}$$

**Q5 (B)**

Given that $\frac{m_1}{m_2}=\frac{4}{3}, \frac{r_1}{r_2}=\frac{3}{4}$

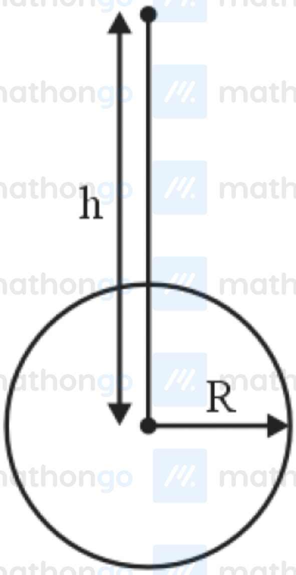
Now $TE=\frac{1}{2}mv^2+\left(\frac{-GMm}{r}\right)$

but $\frac{mv^2}{r}=\frac{GMm}{r^2} \Rightarrow mv^2=\frac{GMm}{r}$

$\Rightarrow TE=-\frac{GMm}{2r} \propto \frac{m}{r}$

$\frac{TE_1}{TE_2}=\frac{m_1}{m_2} \cdot \frac{r_2}{r_1}=\frac{4}{3} \times \frac{4}{3}=\frac{16}{9}$

**Q6 (B)**




$$-\frac{GMm}{R}+\frac{1}{2}m\lambda^2V_e^2=-\frac{GMm}{h}$$

$$-\frac{GMm}{R}+\frac{1}{2}\lambda^2\frac{2GMm}{R}=-\frac{GMm}{h}$$

$$\frac{\lambda^2}{R}-\frac{1}{R}=\frac{-1}{h}$$

$$\frac{1}{h}=\frac{1-\lambda^2}{R}$$

$$h=\frac{R}{1-\lambda^2}$$

**Q7 (D)**

$$g = \frac{GM}{R^2}$$

$$M = \text{constant} \quad g < \frac{1}{R^2}$$

$$100\,\frac{\Delta g}{g} = -2\,\frac{\Delta R}{R}\,100$$

$$\%\ \text{change} = -\,2\,(-2)$$

$$\%\ \text{change in } g = 4\%$$

increase by 4%

**Q8 (C)**



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$$t \propto \frac{1}{\sqrt{g}} \text{ and } g \propto \frac{1}{(R+h)^2}$$

$$\frac{t_1}{t_2} = \sqrt{\frac{g'}{g}} = \sqrt{\frac{R^2}{(R+h)^2}}$$

$$\frac{t_1}{t_2} = \frac{4}{6} = \frac{R}{(R+h)} \Rightarrow h = 3200\,\text{km}$$

**Q9 (2)**

$$g\left(1 - \frac{2h}{R}\right) = g\left(1 - \frac{d}{R}\right)$$

$$\frac{2h}{R} = \frac{d}{R}$$

$$\alpha h = d$$

$$\alpha = 2$$

**Q10 (A)**

$$U_i = \frac{-GMm}{R}$$

$$U_f = -\frac{GMm}{4R}$$

$$\Delta U = U_f - U_i = \frac{3GMm}{4R}$$

$$= \frac{3}{4}mgR$$

$$= \frac{3}{4} \times 1 \times 10 \times 64 \times 10^5$$

$$= 48 \text{ MJ}$$

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